

MIMO ILC with basis functions applied to a flatbed printing system

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Traineeship report

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Abstract

The productivity of printing systems is determined by the amount of medium that can be processed per time unit. The Arizona flatbed printer, developed by Océ-Technologies B.V. in Venlo, The Netherlands, is a printing system suited for high-quality prints on both rigid and flexible media, see Figure 1.

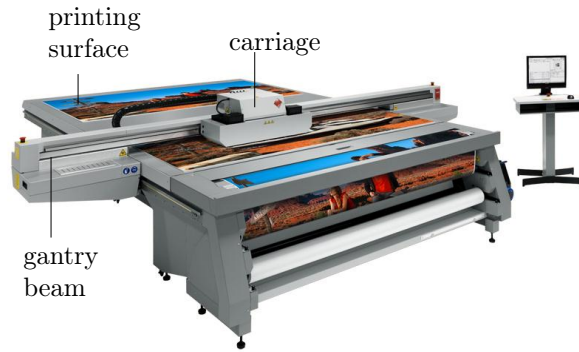


Figure 1: The Arizona flatbed printer

The carriage, with printheads jetting ink onto the medium, performs a point-to-point motion from left-to-right and vice versa, which is called a pass. Between two successive passes, the gantry beam performs a point-to-point motion in longitudinal direction, which is called a step. By repeating this step along the complete length of the surface, the entire printing surface is covered.

The productivity of the Arizona flatbed printer is related to the time required to perform the point-to-point motions of both the gantry and the carriage. To improve the productivity of the printing system more aggressive motion profiles are introduced. The focus of this work is on the point-to-point motion of the gantry beam.

To achieve high tracking performance feed forward control is of crucial importance. It is common practice to manually tune feed forward controllers. However, this is a time-consuming process. Therefore, an auto-tuning feed forward control strategy is desired. To this end, iterative learning control (ILC) with basis functions is considered.

Both a SISO ILC controller as well as a MIMO ILC controller are designed. The accuracy requirements consists of $\pm 5 \mu\text{m}$ error bounds around the end position of the left and right end of the gantry beam. The experimental results show that the designed MIMO ILC controller satisfies the accuracy requirements, in contrast to the SISO ILC controller which only achieves satisfactory performance at the center of the gantry beam.

Since the Arizona printing system has different printing modes, extrapolation properties are essential to guarantee satisfactory performance. Therefore, a set of references is considered consisting of three different motion profiles. Experimental results show that the designed MIMO ILC controller with basis functions achieves good tracking performance for the entire set of motion profiles.

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Chapter 1

Introduction

1.1 Background

The productivity of printing systems is determined by the amount of medium that can be processed per time unit. The Arizona flatbed printer, developed by Océ-Technologies B.V. in Venlo, The Netherlands, is a printing system suited for high-quality prints on both rigid and flexible media, see Figure 1.1.



Figure 1.1: The Arizona flatbed printer

During the printing process both the carriage and the gantry perform point-to-point motions. The productivity of the Arizona flatbed printer is related to the time required to perform such a point-to-point motion. The smaller the time required to perform the motion task, the more products can be processed per time unit. Therefore, reducing the process time by introducing more aggressive motion profiles results directly in an increase in productivity provided that the required accuracy is achieved. This situation can be translated into a tracking control problem. The gantry and the carriage (the plant) are forced to follow an aggressive motion profile (the reference) within specific quality requirements (error bounds).

1.2 Motivation

To track motion profiles with high accuracy, machines are typically equipped with a feedback controller and a feed forward controller. The design of the feed forward controller is of crucial importance to achieve the performance requirements in high-precision applications such as the Arizona flatbed printer [1].

Industrial feed forward controllers are normally tuned manually. This can be a time-consuming process. Furthermore, manually tuned feed forward control performs well only to the extent that the system is accurately known. Friction and unmodeled non-linear behavior can limit the effectiveness of feed forward control [2]. Therefore, data-driven auto-tuning feed forward control is desired.

Iterative learning control (ILC) is a learning-based control methodology for tracking a reference trajectory in repetitive systems. The basic idea behind ILC is to improve the performance of the system by adjusting the input to the system during future trials based on errors measured during previous trials. ILC differs from other learning-type control strategies, such as adaptive control since adaptive control strategies modify the controller, which is a system, whereas ILC modifies the control input, which is a signal.

ILC seems a suitable control strategy for the Arizona flatbed printer. However, ILC shows performance degradation when different tasks have to be performed [3]. Therefore, it is not suitable for a printing system such as the Arizona flatbed printer which has different printing modes and therefore different motion tasks.

This problem may be solved by extending ILC with basis functions [3] and [4]. The basic idea of ILC with basis functions is to learn feed forward parameters and use these parameters to construct a feed forward signal based on the considered motion profile.

1.3 Problem formulation

The focus of this work is on the design and the implementation of ILC with basis function to the gantry of the Arizona flatbed printer. The problem is formulated as follows:

Design and implement data-driven MIMO feed forward control which satisfies the accuracy requirements for a class of aggressive motion profiles.

1.4 Report lay-out

This report is organized as follows. First the Arizona flatbed printer is described in detail in chapter 2. Chapter 3 describes the considered test case scenario and in chapter 4 the iterative learning control theory is refreshed. Then, in chapter 5 the experimental results of the designed ILC controllers are presented and extrapolation properties are discussed. Finally, conclusions and recommendation for future work are presented in chapter 6.

Chapter 2

System description

In this chapter the Arizona flatbed printer as considered in this work is presented. In section 2.1 the Arizona flatbed printer is introduced and the printing process is described. In section 2.2 the considered control system is presented and in section 2.3 a MIMO system identification of the flatbed printer is performed. Section 2.4 introduces the feedback control implemented in the flatbed printer and in section 2.5 the stability of the system is discussed. Finally, in section 2.6 the interaction of the system is analyzed.

2.1 The Arizona flatbed printer

The Arizona flatbed printer is presented in Figure 2.1.

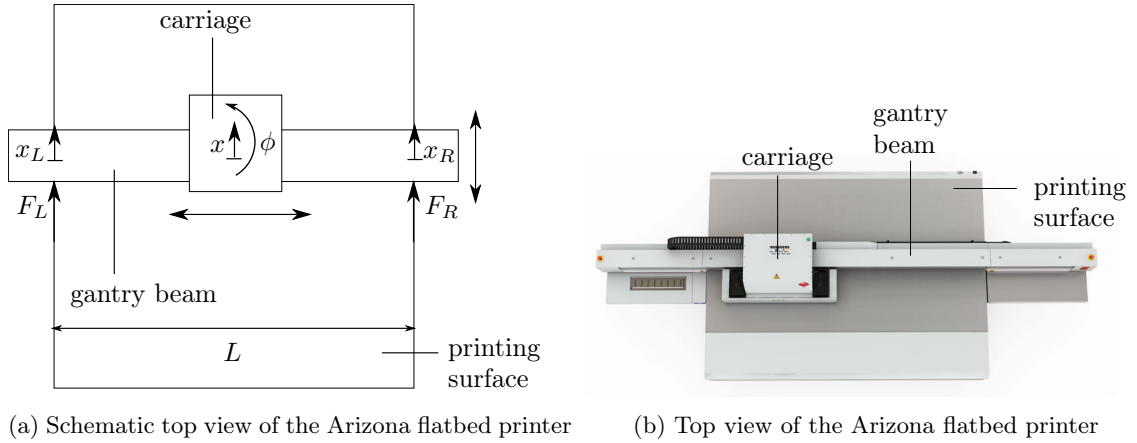


Figure 2.1: The Arizona flatbed printer

The Arizona flatbed printer consists of a series of inkjet printheads which are located in the carriage. The carriage is supported by a gantry beam and moves laterally from left-to-right and vice versa, which is called a pass. The gantry beam is mounted on a guidance which is attached to the printing surface. Between two successive passes, the gantry beam performs a step in longitudinal direction. By repeating this step along the complete length of the surface, the entire printing surface is covered. The gantry beam is actuated by two motors resulting in the forces F_L and F_R . Two linear encoders measure the positions x_L and x_R of the gantry beam. As a result, the gantry beam of the Arizona flatbed printer can be considered as a 2x2 MIMO system with inputs F_L and F_R and outputs x_L and x_R .

2.2 Control system

The control system applied to the gantry beam of the Arizona flatbed printer is presented in Figure 2.2. The control system consists of the 2x2 MIMO plant P which represent the gantry beam of the

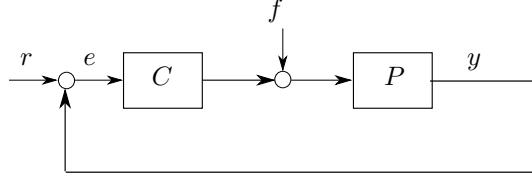


Figure 2.2: The control system applied to the gantry beam of the Arizona flatbed printer.

Arizona flatbed printer. The plant P is forced to follow the reference $r \in \mathbb{R}^{2 \times 1}$. A feedback controller C is used to stabilize the plant and achieve disturbance rejection and a feed forward signal $f \in \mathbb{R}^{2 \times 1}$ is applied to improve tracking performance.

In order to control the system kinematic relations are used to decouple the system into the average position x and the rotation ϕ . The average position x is defined as:

$$x = \frac{1}{2} \cdot (x_R + x_L) \quad (2.1)$$

and the rotation ϕ is defined as:

$$\phi = \text{atan} \left(\frac{360}{2\pi} \frac{1}{L} \cdot (x_R - x_L) \right) \quad (2.2)$$

with L the length of the gantry beam from left encoder to right encoder.

2.3 System identification

In order to identify the MIMO system, first the relevant MIMO transfers are derived. Consider the control system of Figure 2.3.

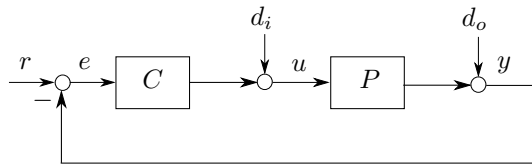


Figure 2.3: Feedback control system with disturbance at the input d_i and disturbance at the output d_o .

The input loop transfer matrix, L_i , and the output loop transfer matrix L_o are defined as

$$L_i = CP \quad (2.3)$$

$$L_o = PC \quad (2.4)$$

where L_i is obtained from breaking the loop at the input (u) of the plant while L_o is obtained from breaking the loop at the output (y) of the plant.

The input sensitivity matrix S_i is defined as the transfer matrix from d_i to u :

$$S_i = (I + L_i)^{-1}, u = S_i d_i \quad (2.5)$$

and the output sensitivity matrix S_o is defined as the transfer matrix from d_o to y :

$$S_o = (I + L_o)^{-1}, y = S_o d_o. \quad (2.6)$$

From Figure 2.3 it follows that:

$$e = S_o r - S_o P d_i - S_o d_o. \quad (2.7)$$

For SISO systems, it is common practice to measure d_i , u and e in order to obtain S and $-PS$. Using S and $-PS$ the plant is then identified by

$$P = \frac{PS}{S}. \quad (2.8)$$

However, for MIMO systems the choice of the sensitivity function is of crucial importance. During closed-loop identification noise is injected at the input of the plant. Therefore, the input sensitivity should be used. However, (2.7) shows that the transfer matrix from d_i to e is given by $-S_o P$, which depends on S_o instead of S_i . To solve this problem, the *push-through rule* is used:

Push-through rule

If $(I + AB)$ and $(I + BA)$ are non-singular, then

$$A(I + BA)^{-1} = (I + AB)^{-1}A. \quad (2.9)$$

For $A = P$ and $B = C$, the push-through rule results in

$$P(I + CP)^{-1} = (I + PC)^{-1}P \quad (2.10)$$

$$PS_i = S_o P. \quad (2.11)$$

Using (2.11) in (2.7) results in

$$e = S_o r - PS_i d_i - S_o d_o. \quad (2.12)$$

The transfer matrices S_i and $-PS_i$ are obtained by measuring d_i , u and e . The MIMO plant P is then identified using

$$P = PS_i S_i^{-1}. \quad (2.13)$$

The MIMO FRF measurements and the MIMO model fit are plotted in Figure 2.4.

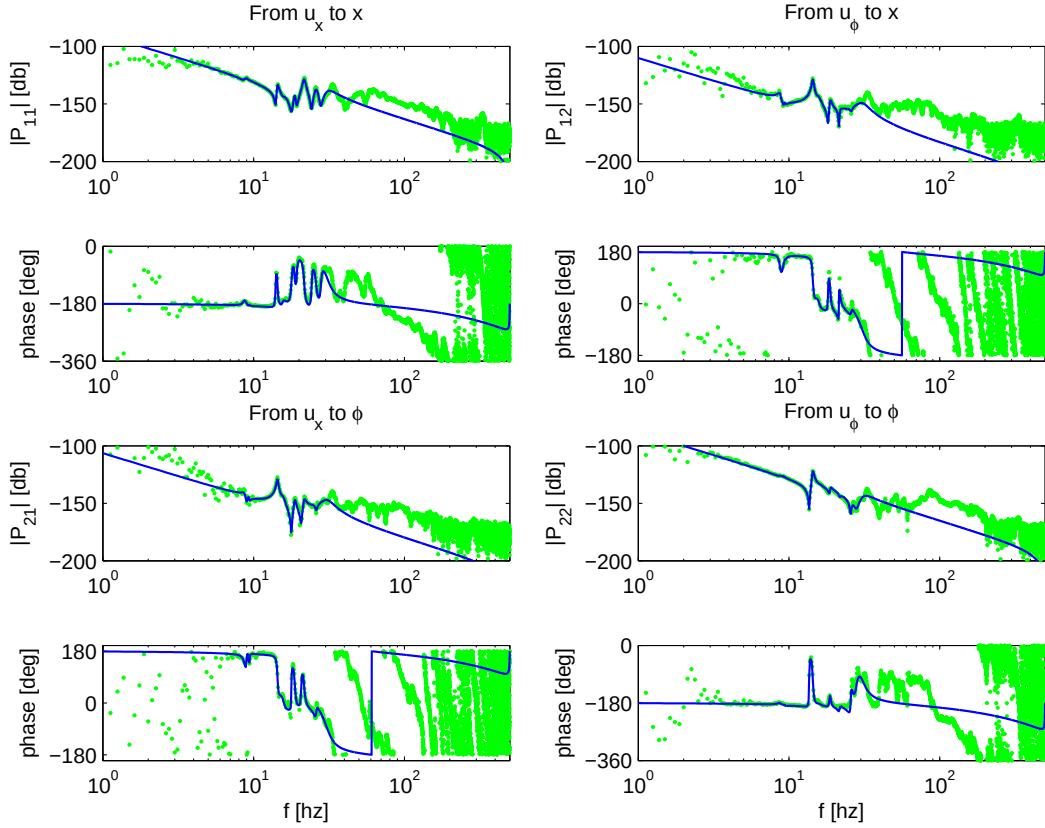


Figure 2.4: MIMO system P with inputs u_x , u_ϕ and outputs x , ϕ : frequency response measurement (dotted), model fit (solid).

In order to understand the dynamic behavior of the gantry beam of the flatbed printer, a schematic side view of the flatbed printer is presented in Figure 2.5 [9].

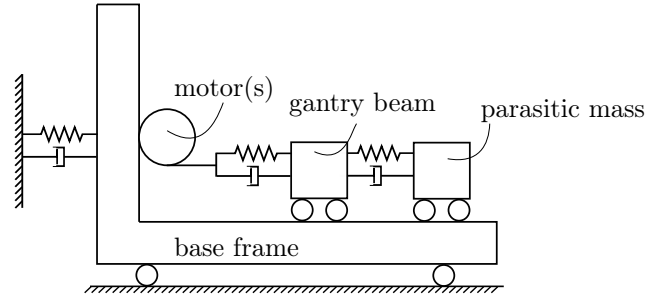


Figure 2.5: Schematic side view of the Arizona flatbed printer.

The flatbed printer is assumed to consist of a base frame which is attached to the fixed world. The motor(s) which actuate the gantry beam are connected to the base frame. The carriage is considered as a parasitic mass and is connected to the gantry beam.

The resonances and anti-resonances mostly concentrated between 10 and 50 Hz shown in Figure 2.4 are typically due to the flexible connection between the fixed world and base frame and the flexible connection between the gantry beam and parasitic masses. The resonance frequency at 130 Hz is due to the flexibility between the motor(s) and the gantry beam. This flexibility typically introduced phase-lag, see Figure 2.4.

2.4 Feedback control

In order to stabilize the system and achieve disturbance rejection a diagonal feedback controller C is present in the printer. The feedback controller is presented in Figure 2.6 and the open loop transfer is presented in Figure 2.7.

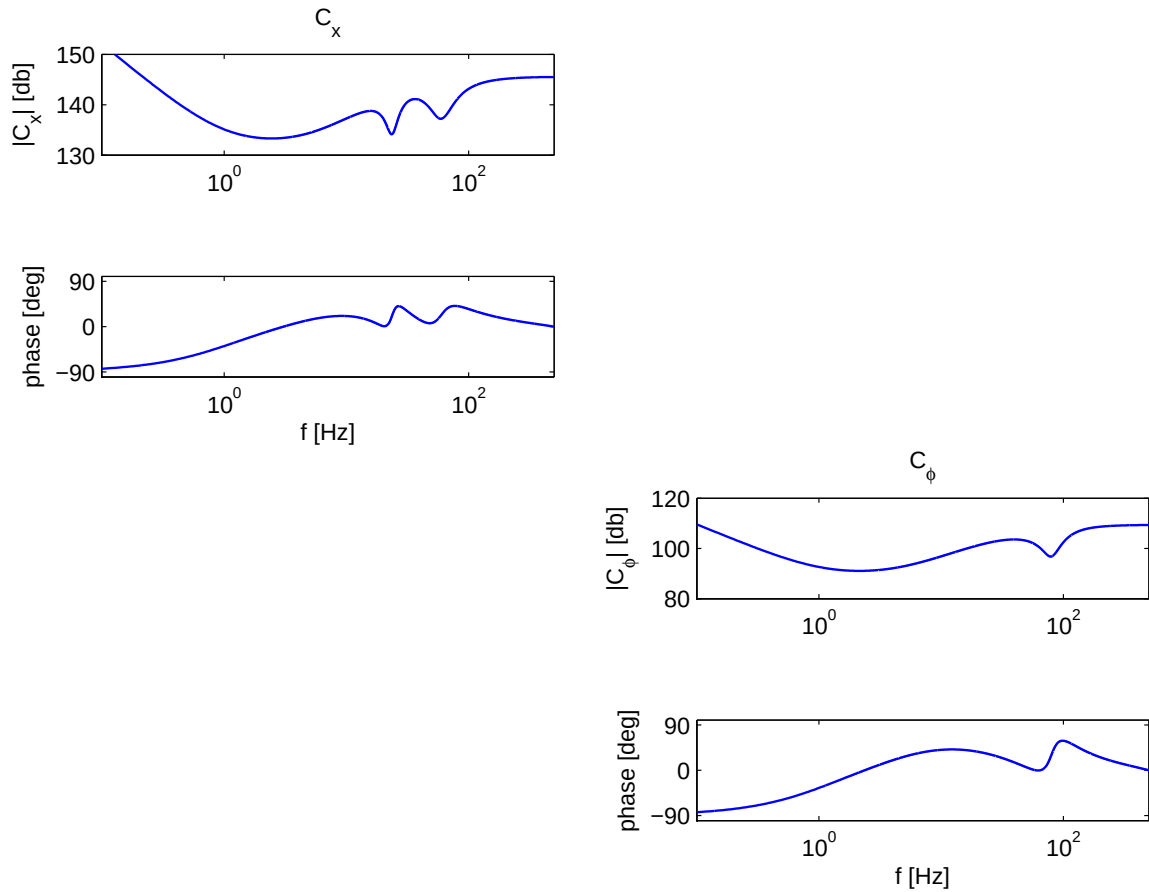


Figure 2.6: The feedback controller implemented in the Arizona flatbed printer.

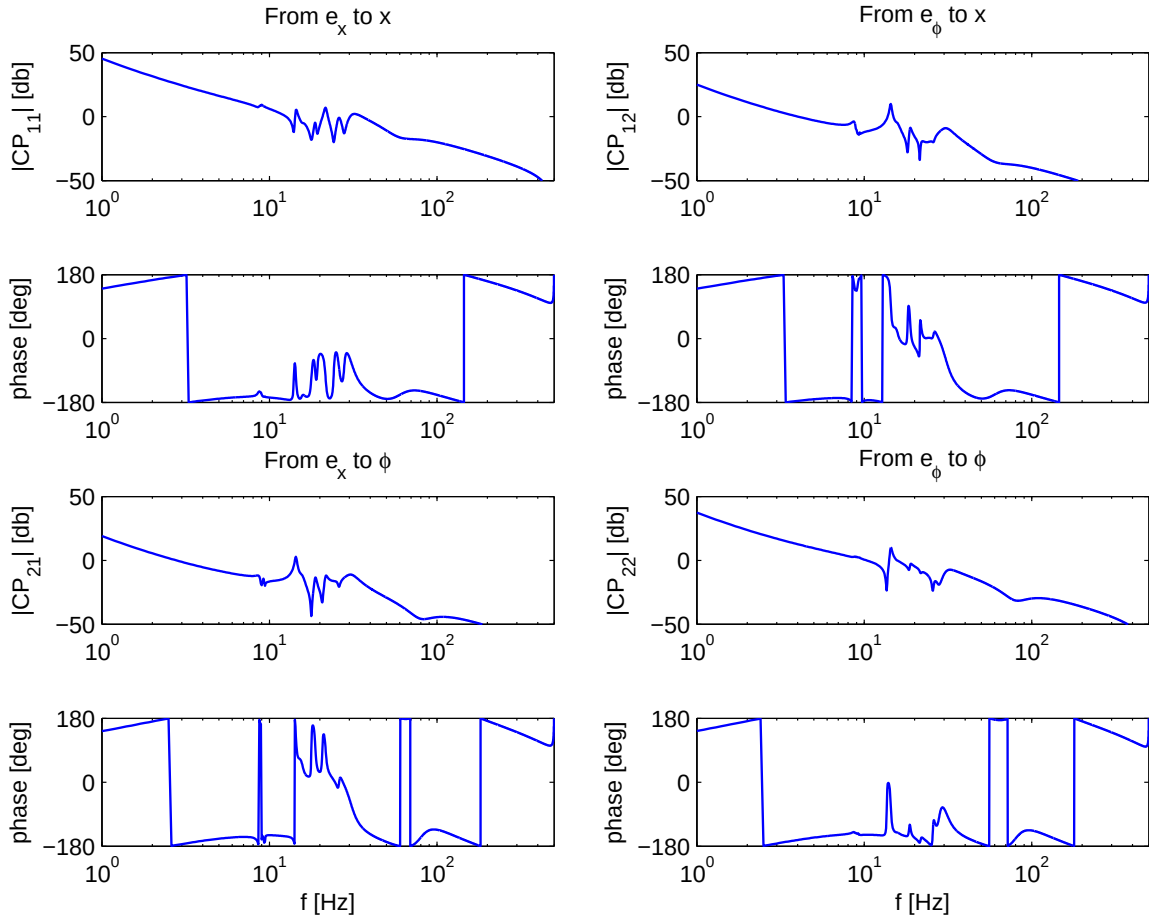


Figure 2.7: The open-loop transfer CP .

The first diagonal element C_x corresponding to x consists of an integrator, a lead-lag filter and two Notch filters. The second diagonal element C_ϕ corresponding to ϕ consists of an integrator, a lead-lag filter and a Notch filter.

2.5 Stability

Stability of multivariable systems is governed by the Generalized Nyquist Criterion. The characteristic loci λ_i of the transfer function $P(s)$ are defined to be the eigenvalues of $P(j\omega_i)$ plotted per frequency ω_i . For these characteristic loci the following theorem holds:

Generalized Nyquist Criterion

$G(s)$ has p_0 unstable (Smith-McMillan) poles, then the closed-loop system with return ratio $-kG(s)$ is stable if and only if the characteristic loci of $kG(s)$, taken together, encircle the point -1 p_0 times anti-clockwise, assuming that there are no hidden modes.

The considered system is a 2x2 MIMO system. Therefore, at each frequency ω_i the system has two eigenvalues. As a results, two characteristic loci exist. The characteristic loci of the considered MIMO system P are plotted in Figure 2.8. Both loci encircle the point -1 anti-clockwise.

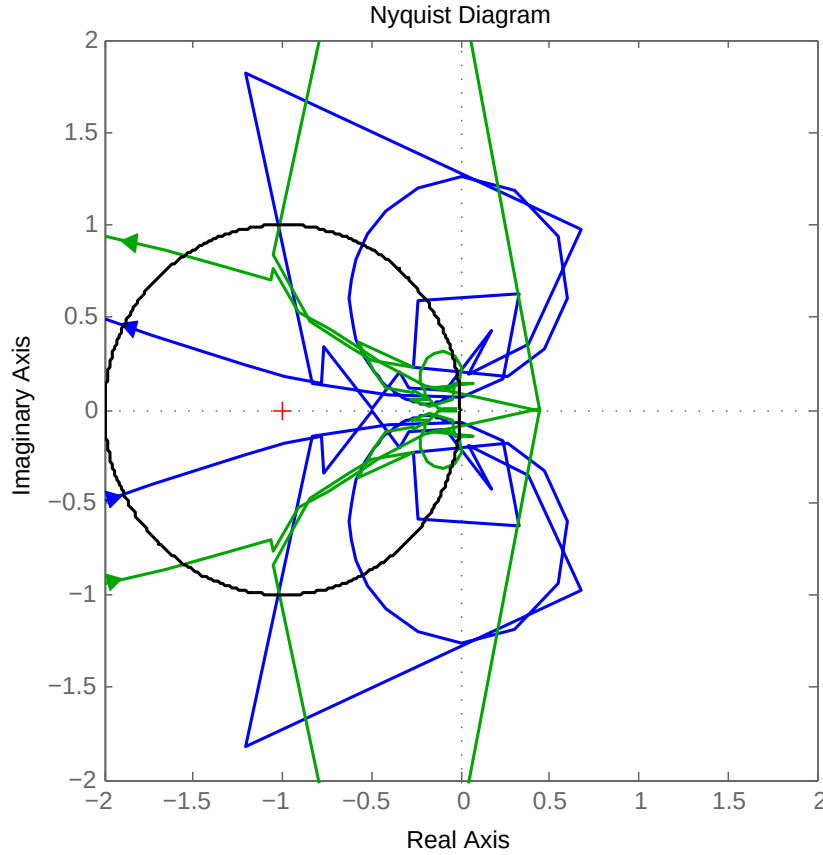


Figure 2.8: The characteristic loci of the MIMO system P .

Therefore, it can be concluded that the system is stable. However, it should be noted that poor stability margins are achieved. This is also confirmed by Figure 2.9 where the MIMO sensitivity function is plotted.

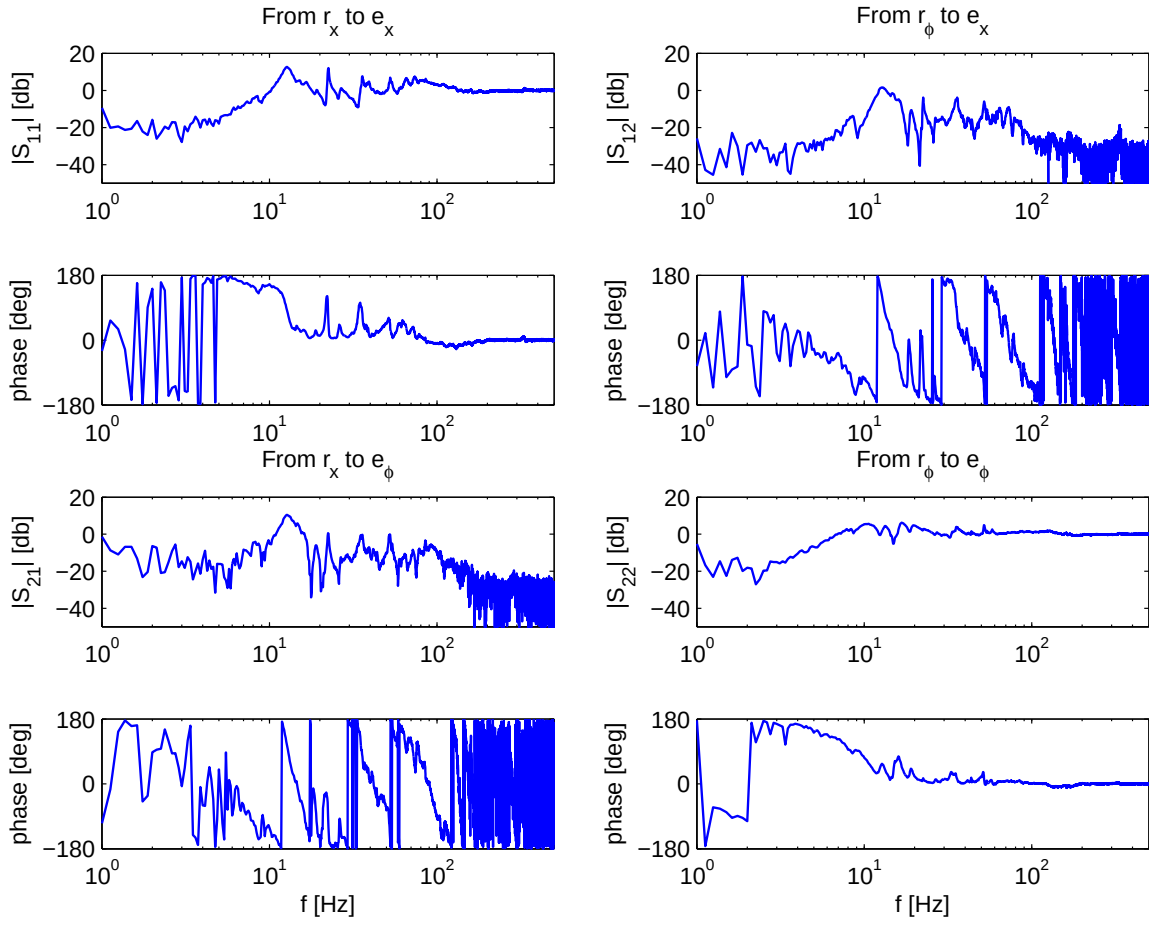


Figure 2.9: The sensitivity function S of the MIMO system P .

Large sensitivity peaks occur for S_{11} (12 db at 12.4 Hz and 22.5 Hz) and S_{21} (10 db at 13 Hz) . Therefore, it can be concluded that there is still room for improvement in the feedback control.

2.6 Interaction

The interaction index ϕ [10] of a matrix G is defined as

$$\phi = \frac{g_{12}g_{21}}{g_{11}g_{22}}. \quad (2.14)$$

When ϕ is close to zero this means no interaction. Figure 2.10 shows ϕ as a function of frequency for the MIMO system P .

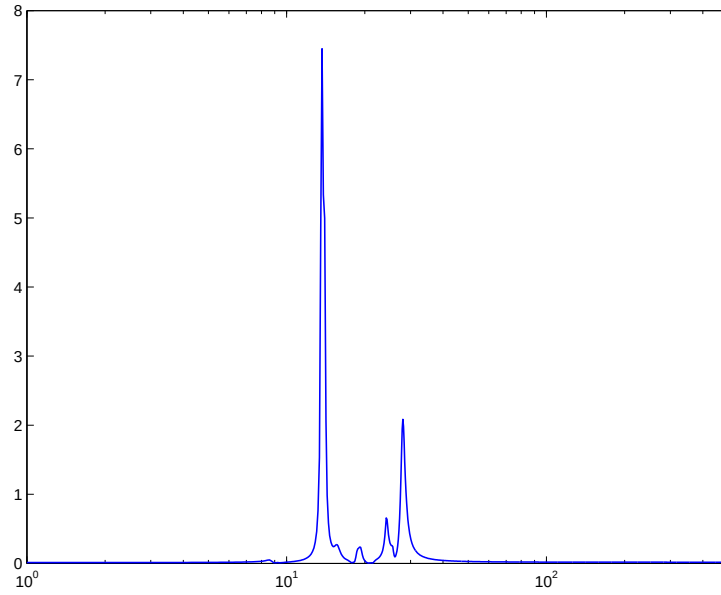


Figure 2.10: The interaction index of the MIMO system P as a function of frequency.

In the region where rigid body is dominant, up to 10 Hz, there is no significant interaction. However, in the region where flexible dynamics is dominant there is significant interaction. Therefore, to achieve high-tracking performance it is crucial to consider the MIMO characteristics of the system.

Chapter 3

Test scenario

In this chapter the considered test scenario is described. In section 3.1 the considered motion profiles are introduced and the accuracy requirements for the printing process are described in section 3.2.

3.1 Motion profiles

An important motion profile for the gantry beam of the flatbed printer consists of a step of 41 mm which is equivalent to the width of one printhead.

Normally, the 41-mm-step is performed in 0.8 s with a maximum acceleration of 0.55 m/s^2 . To increase the productivity, motion profiles are considered with a maximum acceleration of 0.9 m/s^2 . This maximum acceleration is a constraint due to the properties of the electric motors used for the gantry beam. For accelerations higher than 0.9 m/s^2 , the motors rapidly heat up resulting in performance degradation.

The acceleration profile a_{41} for the 41-mm-step with a maximum acceleration of 0.9 m/s^2 is presented in Figure 3.1a. From this acceleration profile the motion profile is constructed by integrating twice the acceleration profile. The resulting 41-mm-step is performed in 0.6 s. For evaluation purposes an additional 200 ms are added to the motion profile to investigate the settling behavior of the gantry beam. The motion profile r_{41} corresponding to the 41-mm-step is plotted in Figure 3.1b.

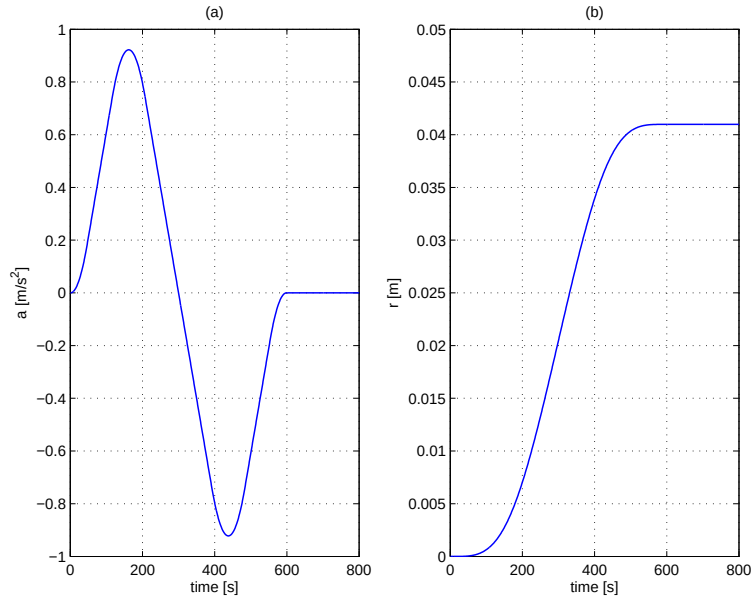


Figure 3.1: (a): The acceleration profile a_{41} . (b): The motion profile r_{41} .

In order to investigate extrapolation properties a set of references is considered. The set consists of the previously introduced r_{41} , an 20-mm-step and an 61-mm-step. The 20-mm-step and the 61-mm-step are referred to as r_{20} and r_{61} respectively.

The motion profiles r_{20} and r_{61} are constructed using the same maximum acceleration as for r_{41} . The difference is the duration of the acceleration. Motion profile r_{20} is constructed using half of the time of a_{41} and r_{61} is constructed using 1.5 the time of a_{41} .

The acceleration profiles a_{20} , a_{41} and a_{61} and the motion profiles r_{20} , r_{41} and r_{61} are plotted in Figure 3.2.

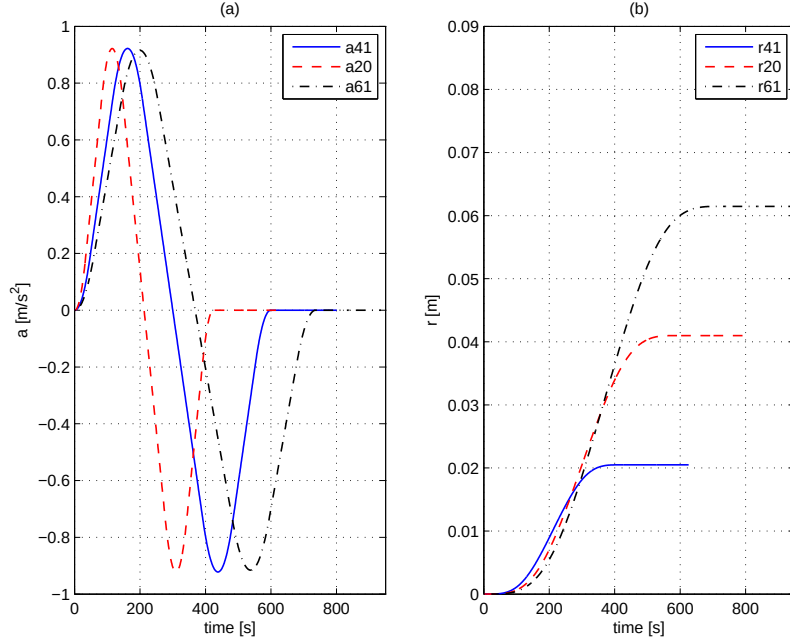


Figure 3.2: (a). The acceleration profiles a_{20} , a_{41} and a_{61} . (b). The motion profiles r_{20} , r_{41} and r_{61} .

3.2 Accuracy requirements for the printing process

The gantry beam performs a point-to-point motion. Therefore, the required positioning accuracy for the printing process is specified by $\pm 5 \mu\text{m}$ around the final value of the reference. The positioning accuracy should be satisfied along the entire length of the gantry beam, since the position of the carriage varies during the printing task. Therefore, the accuracy requirements can be defined as follows:

$$-5 \mu\text{m} < e_{x_L} < 5 \mu\text{m} \quad (3.1)$$

$$-5 \mu\text{m} < e_{x_R} < 5 \mu\text{m} \quad (3.2)$$

where

$$\begin{aligned} e_{x_L} &= e_x - \frac{L}{2} \sin\left(\frac{\pi}{180} e_\phi\right) \\ e_{x_R} &= e_x + \frac{L}{2} \sin\left(\frac{\pi}{180} e_\phi\right) \end{aligned} \quad (3.3)$$

with e_{x_L} the error at the left end of the gantry beam and e_{x_R} the error at the right end of the gantry beam. The goal is to achieve zero settling time. Therefore (3.3) should be satisfied during the last 200 ms of a trial.

Chapter 4

Iterative learning control

In this chapter the theory related to iterative learning control is revisited. In section 4.1. the lifted system representation is introduced for both SISO and MIMO systems. ILC with basis functions for SISO systems is treated section 4.2. Performance, stability and convergence and discussed in section 4.3, 4.4 and 4.5 respectively. In section 4.6 the framework introduced in section 4.2 is extended to MIMO systems. Finally, in section 4.7 the feed forward parametrization is discussed.

4.1 Lifted system representation

Consider the SISO LTI discrete-time system $H(z)$. The system behavior on the finite time interval $k = 0, 1, \dots, N - 1$, with N the total number of samples is represented by the convolution matrix $\mathbf{H} \in \mathbb{R}^{N \times N}$. It contains the impulse response data $h(k)$ for time $k = 0, 1, \dots, N - 1$:

$$\mathbf{H} = \begin{bmatrix} h(0) & 0 & \cdots & 0 \\ h(1) & h(0) & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ h(N-1) & \cdots & h(1) & h(0) \end{bmatrix}. \quad (4.1)$$

The systems input vector $\mathbf{u} \in \mathbb{R}^{N \times 1}$ and output vector $\mathbf{y} \in \mathbb{R}^{N \times 1}$ are defined as:

$$\mathbf{u} = \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(N-1) \end{bmatrix}. \quad (4.2)$$

In this way the system can be expressed in the form

$$\mathbf{y} = \mathbf{H}\mathbf{u}. \quad (4.3)$$

For a 2x2 MIMO LTI discrete-time system $G(z)$, the impulse response $g(k)$ contains the impulse response from each input to each output resulting in the following convolution matrix $\mathbf{G}$:

$$\mathbf{G} = \begin{bmatrix} g_{11}(0) & g_{12}(0) & 0 & 0 & \cdots & 0 & 0 \\ g_{21}(0) & g_{22}(0) & 0 & 0 & \cdots & 0 & 0 \\ g_{11}(1) & g_{12}(1) & g_{11}(0) & g_{12}(0) & \cdots & \vdots & \vdots \\ g_{21}(1) & g_{22}(1) & g_{21}(0) & g_{22}(0) & \cdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ g_{11}(N-1) & g_{12}(N-1) & \cdots & \cdots & \cdots & g_{11}(0) & g_{12}(0) \\ g_{21}(N-1) & g_{22}(N-1) & \cdots & \cdots & \cdots & g_{21}(0) & g_{22}(0) \end{bmatrix}. \quad (4.4)$$

For a 2x2 MIMO systems the input vector $\mathbf{u} \in \mathbb{R}^{2N \times 1}$ and output vector $\mathbf{y} \in \mathbb{R}^{2N \times 1}$ are defined as:

$$\mathbf{u} = \begin{bmatrix} u_1(0) \\ u_2(0) \\ u_1(1) \\ u_2(1) \\ \vdots \\ u_1(N-1) \\ u_2(N-1) \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1(0) \\ y_2(0) \\ y_1(1) \\ y_2(1) \\ \vdots \\ y_1(N-1) \\ y_2(N-1) \end{bmatrix}. \quad (4.5)$$

In this way the system can be represented by

$$\mathbf{y} = \mathbf{G}\mathbf{u}. \quad (4.6)$$

4.2 SISO ILC with basis functions

Consider the feedback control system with feed forward control of Figure 4.1.

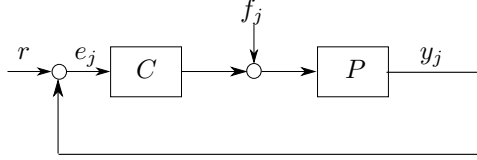


Figure 4.1: Feedback control system with feed forward control.

In ILC with basis functions the feed forward signal $f_j \in \mathbb{R}^{N \times 1}$ at trial j is constructed from a linear combination of user-defined vectors $\Psi = [\psi_1, \psi_2, \dots, \psi_m] \in \mathbb{R}^{N \times m}$ and feed forward parameters $\theta_j \in \mathbb{R}^{m \times 1}$:

$$f_j = \Psi \theta_j \quad (4.7)$$

with m the number of user-defined vectors $\psi \in \mathbb{R}^{N \times 1}$ and feed forward parameters θ_j .

The ability to track different reference trajectories is achieved by choosing the user-defined vectors as a function of the reference trajectory r :

$$f_j = \Psi(r) \theta_j. \quad (4.8)$$

Once, the user-defined vectors ψ are chosen, the parameters θ_j are 'learned' through data-driven practice by solving an optimization problem.

Calculating the ILC feed forward update f_{j+1} by solving an optimization problem is referred to as Norm-optimal ILC [6], [7]. The optimization criterion for norm-optimal ILC with basis functions is given by

$$J(\theta_{j+1}) := \|e_{j+1}\|_{W_e} + \|f_{j+1}\|_{W_f} + \|f_{j+1} - f_j\|_{W_{\Delta f}} \quad (4.9)$$

where e_{j+1} can be written as

$$e_{j+1} = e_j - SP\Psi(\theta_{j+1} - \theta_j) \quad (4.10)$$

and the weighted 2-norm is defined as

$$\|x\|_W = x^T W x \quad (4.11)$$

with $W \succ 0 \in \mathbb{R}^{N \times N}$ the weighting matrix.

Since f_j is linear in the parameters θ_j , $J(\theta_{j+1})$ results in a quadratic optimization criterion. The solution of this optimization criterion can be computed analytically and is obtained from the necessary condition for optimality

$$\frac{\partial J(\theta_{j+1})}{\partial \theta_{j+1}} = 0. \quad (4.12)$$

Solving for θ_{j+1} results in

$$\begin{aligned} \theta_{j+1} &= L e_j + Q \theta_j, \\ L &= [\Psi^T (S_p^T W_e S_p + W_f + W_{\Delta f}) \Psi]^{-1} [\Psi^T S_p^T W_e], \\ Q &= [\Psi^T (S_p^T W_e S_p + W_f + W_{\Delta f}) \Psi]^{-1} [\Psi^T (S_p^T W_e S_p + W_{\Delta f}) \Psi] \end{aligned} \quad (4.13)$$

with $L \in \mathbb{R}^{N \times N}$ and $Q \in \mathbb{R}^{N \times N}$ the learning matrices, $\theta_{j+1} \in \mathbb{R}^{m \times 1}$ the parameter update and $S_p := S P \in \mathbb{R}^{N \times N}$ the process sensitivity in lifted form. In (4.13), $W_e \succ 0 \in \mathbb{R}^{N \times N}$, $W_f \in \mathbb{R}^{N \times N}$ and $W_{\Delta f} \succeq 0 \in \mathbb{R}^{N \times N}$ are used-defined weighting matrices to specify performance and robustness objectives, namely, robustness with respect to model uncertainty (W_f) and convergence speed and sensitivity to trial varying disturbances ($W_{\Delta f}$). Based on θ_{j+1} the feed forward update is defined

$$f_{j+1} = \Psi \theta_{j+1}. \quad (4.14)$$

4.3 Performance

The tracking error $e_j \in \mathbb{R}^{N \times 1}$ can be derived from Figure 4.1

$$e_j = S r - S_p f_j. \quad (4.15)$$

Equation (4.15) shows that choosing $f_j = P^{-1} r$ results in $e_j = 0$. As argued in 4.2, in case of Norm-optimal ILC with basis functions the best approximation of $P^{-1} r$ is computed by minimizing (4.9). Therefore, (4.9) can be seen a measure for tracking performance. Setting the used-defined matrices $W_f = 0$ and $W_{\Delta f} = 0$ results in the following definition of tracking performance

$$J = e^T W_e e. \quad (4.16)$$

4.4 Stability

The dynamics of the feed forward parameters θ can be derived using (4.13) and (4.15) resulting in

$$\theta_{j+1} = (Q - L S_p \Psi) \theta_j + L S r. \quad (4.17)$$

The ILC-controller with basis functions is stable if and only if

$$|\lambda_i(Q - L S_p \Psi)| < 1, i = 1, 2, \dots, N \quad (4.18)$$

where λ_i denotes the eigenvalues. Since f is defined as $f = \Psi \theta$ with Ψ constant for a given reference trajectory, stability is guaranteed when (4.18) is satisfied.

4.5 Convergence

The system with ILC controller is monotonic convergent under a given norm $\|\bullet\|$ if

$$\|e_\infty - e_{j+1}\| \leq \gamma \|e_\infty - e_j\| \quad (4.19)$$

for $\gamma \in [0, 1)$.

The error e_j at trial j is defined as

$$e_j = Sr - S_p \Psi \theta_j. \quad (4.20)$$

and the error update e_{j+1} at trial $j + 1$ is defined as

$$e_{j+1} = Sr - S_p \Psi \theta_{j+1}. \quad (4.21)$$

The feed forward parameters θ_∞ for $j \rightarrow \infty$ is defined as $\theta_\infty = \theta_{j+1} = \theta_j$. Using this condition into (4.17) results in

$$\theta_\infty = (Q - LS_p \Psi) \theta_\infty + LSr \quad (4.22)$$

from which the error e_∞ for $j \rightarrow \infty$ can be derived

$$e_\infty = Sr - S_p \Psi \theta_\infty. \quad (4.23)$$

Using (4.20), (4.21) and (4.23) into (4.19) leads to

$$\|e_\infty - e_{j+1}\| \leq \|(Q - LS_p \Psi)\| \|e_\infty - e_j\|. \quad (4.24)$$

Therefore, it can be concluded that monotonic convergence occurs when

$$\|(Q - LS_p \Psi)\| < 1. \quad (4.25)$$

4.6 MIMO ILC with basis functions

In section 4.2 ILC with basis functions was introduced for SISO systems. For the considered 2x2 MIMO system the same approach can be followed.

The feed forward signal $f_j \in \mathbb{R}^{2N \times 1}$ at trial j is constructed from a linear combination of used-defined vectors $\Psi = [\Psi_x, \Psi_\phi] \in \mathbb{R}^{2N \times m+n}$ with $\Psi_x \in \mathbb{R}^{2N \times m}$ and $\Psi_\phi \in \mathbb{R}^{2N \times n}$, and feed forward parameters $\theta_j \in \mathbb{R}^{m+n \times 1}$:

$$f_j = \Psi \theta_j. \quad (4.26)$$

The feed forward signal f_j , the basis Ψ and the feed forward parameters θ_j are defined

$$f_j = \begin{bmatrix} f_{j,x}(0) \\ f_{j,\phi}(0) \\ f_{j,x}(1) \\ f_{j,\phi}(1) \\ \vdots \\ f_{j,x}(N-1) \\ f_{j,\phi}(N-1) \end{bmatrix}, \theta_j = \begin{bmatrix} \theta_{j,x,1} \\ \theta_{j,x,2} \\ \vdots \\ \theta_{j,x,m} \\ \theta_{j,\phi,1} \\ \theta_{j,\phi,2} \\ \vdots \\ \theta_{j,\phi,n} \end{bmatrix}, \quad (4.27)$$

$$\Psi = \begin{bmatrix} \Psi_{x,1}(0) & \Psi_{x,2}(0) & \cdots & \Psi_{x,m}(0) & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \Psi_{\phi,1}(0) & \Psi_{\phi,2}(0) & \cdots & \Psi_{\phi,n}(0) \\ \Psi_{x,1}(1) & \Psi_{x,2}(1) & \cdots & \Psi_{x,m}(1) & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \Psi_{\phi,1}(1) & \Psi_{\phi,2}(1) & \cdots & \Psi_{\phi,n}(1) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \Psi_{x,1}(N) & \Psi_{x,2}(N) & \cdots & \Psi_{x,m}(N) & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \Psi_{\phi,1}(N) & \Psi_{\phi,2}(N) & \cdots & \Psi_{\phi,n}(N) \end{bmatrix}.$$

As a results, the learning matrices $L \in \mathbb{R}^{2N \times 2N}$ and $Q \in \mathbb{R}^{2N \times 2N}$, the process sensitivity $S_p \in \mathbb{R}^{2N \times 2N}$ and the used-defined weighting filters, $W_e \succ 0 \in \mathbb{R}^{2N \times 2N}$, $W_f \in \mathbb{R}^{2N \times 2N}$ and $W_{\Delta f} \succeq 0 \in \mathbb{R}^{2N \times 2N}$. Note that (4.27) is a result of constructing f_x with Ψ_x and f_ϕ with Ψ_ϕ .

In contrast with SISO ILC, for MIMO ILC the error e consists of both e_x and e_ϕ which have different units. To this end, W_e is defined as

$$W_e = \begin{bmatrix} w_{e,x} & 0 & 0 & \cdots & 0 \\ 0 & w_{e,\phi} & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & w_{e,x} & 0 \\ 0 & \cdots & 0 & 0 & w_{e,\phi} \end{bmatrix}. \quad (4.28)$$

To compensate for the difference in units between e_x and e_ϕ , e_ϕ is converted from degrees to meters. The relation between e_ϕ and the resulting error at the end of the gantry beam is given by (3.3). Therefore, $w_{e,x}$ and $w_{e,\phi}$ are defined

$$w_{e,x} = 1 \quad (4.29)$$

$$w_{e,\phi} = \frac{L}{2} \frac{\pi}{180}. \quad (4.30)$$

4.7 Feed forward parametrization

As argued in section 4.1 a control algorithm is required with the ability to track different references. The previous section showed that the ideal feed forward signal $f_j = P^{-1}r$, which depends on the reference trajectory r . Therefore, it is necessary to introduce a feed forward parametrization that incorporates extrapolation capabilities with respect to the reference. This is achieved by choosing basis functions which depend on the reference.

Figure 4.2 shows P^{-1} . All four the entries of P^{-1} show a slope +2 for low frequencies. Therefore, at least the acceleration profile a should be chosen as a basis function. However, it is clear that using only the acceleration profile as a basis function does not result in

$$f_j = \Psi(r)\theta_j = P^{-1}r. \quad (4.31)$$

In order to better approximate $P^{-1}r$, the following extra basis functions are considered

1. The velocity profile v .
2. The reference profile r .
3. The jerk profile j .
4. The snap profile s .
5. A friction compensation c .

The normalized basis functions for reference r_{41} are shown in Figure 4.3. Note that r , v , a , j and s depend on the considered motion profile.

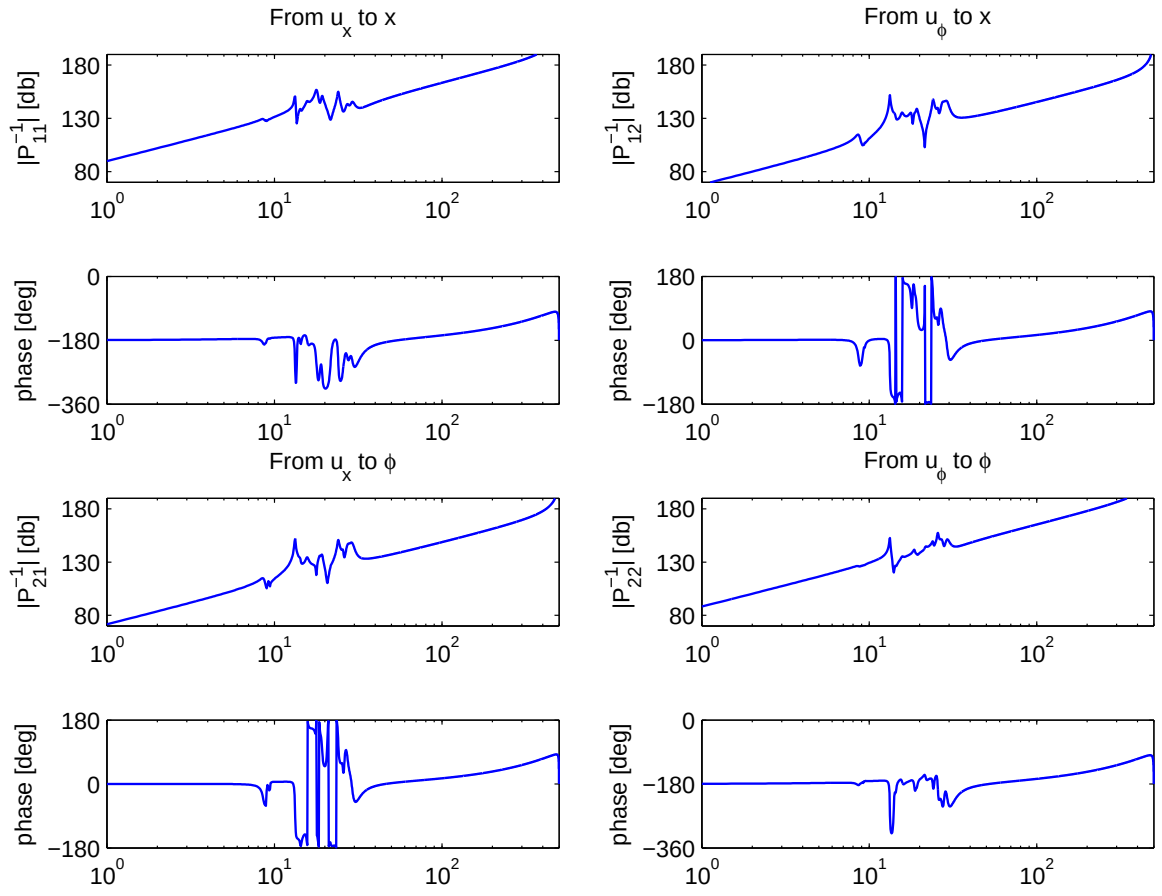


Figure 4.2: P^{-1} of the considered 2x2 MIMO system.

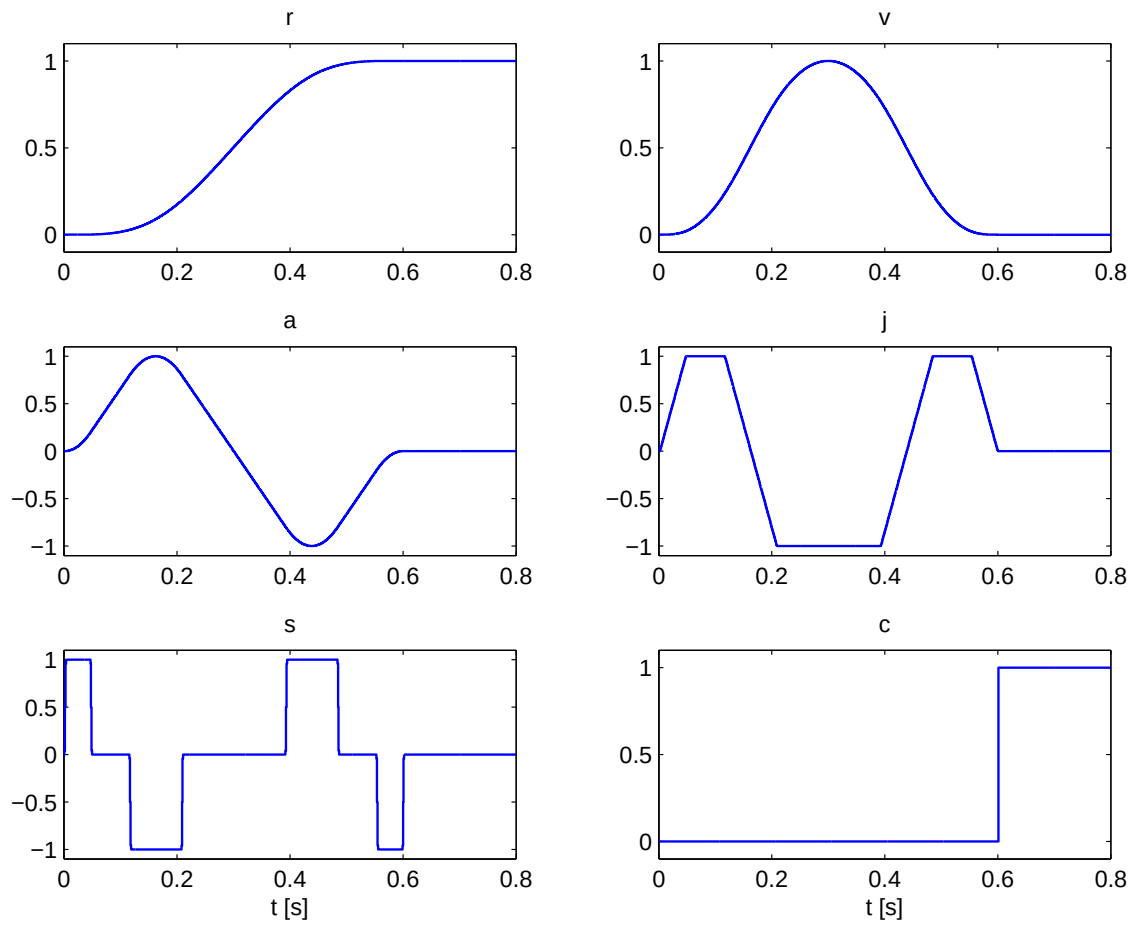


Figure 4.3: The normalized basis functions r , v , a , j , s , c for r_{41} .

Chapter 5

Experimental results

In this chapter the experimental results are presented. The experiments are performed with two different ILC controllers: a SISO ILC controller and a MIMO ILC controller.

The SISO ILC controller constructs a feed forward signal to control the average position x while the MIMO ILC controller computes two feed forward signals: one for the average position x and one for the rotation ϕ .

The performance is investigated by performing 20 successive trials on reference r_{41} and measuring the error e_x on the average position x , the error e_ϕ on the rotation ϕ and computing the positioning error at the left and right end of the beam using (3.3).

The results related to the SISO ILC controller are presented in section 5.1 and the results related to the MIMO ILC controller are presented in section 5.2. Finally, in section 5.3 the extrapolation properties are discussed.

5.1 SISO ILC controller

In order to design a SISO ILC controller, the SISO transfer P_{xx} from u_x to x is identified using a three point identification procedure and a SISO model fit is obtained. The measurements and the model fit are plotted in Figure 5.1.

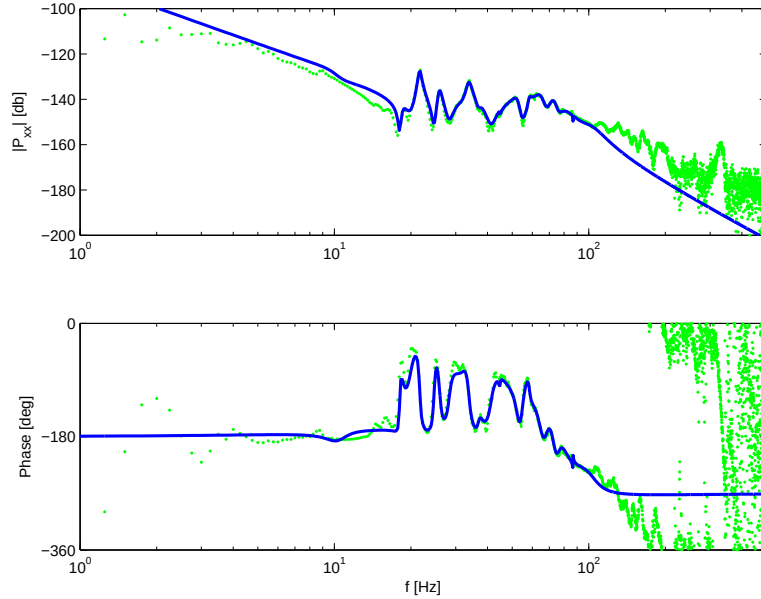


Figure 5.1: SISO transfer P_{xx} from input u_x to output x : frequency response measurement (dotted), model fit (solid).

It should be noted that $P_{xx} \neq P_{11}$ from Figure 2.4. The MIMO closed-loop system with all loops closed is called the equivalent plant and is defined by

$$P_{11}^{eq} = P_{11} - \frac{P_{12}C_\phi P_{21}}{1 + P_{22}C_\phi} \quad (5.1)$$

$$P_{22}^{eq} = P_{22} - \frac{P_{21}C_x P_{12}}{1 + P_{11}C_x}. \quad (5.2)$$

When a closed-loop SISO identification is performed on a MIMO closed-loop system P_{11}^{eq} or P_{22}^{eq} are measured. Therefore, the identified SISO transfer P_{xx} is equivalent to P_{11}^{eq} and since $C_\phi \neq 0$ it follows that $P_{xx} \neq P_{11}$.

The tracking error e_x in the average position x for the SISO ILC controller is depicted in Figure 5.2 where the basis Ψ is extended step wise with the basis functions presented in section 4.7.

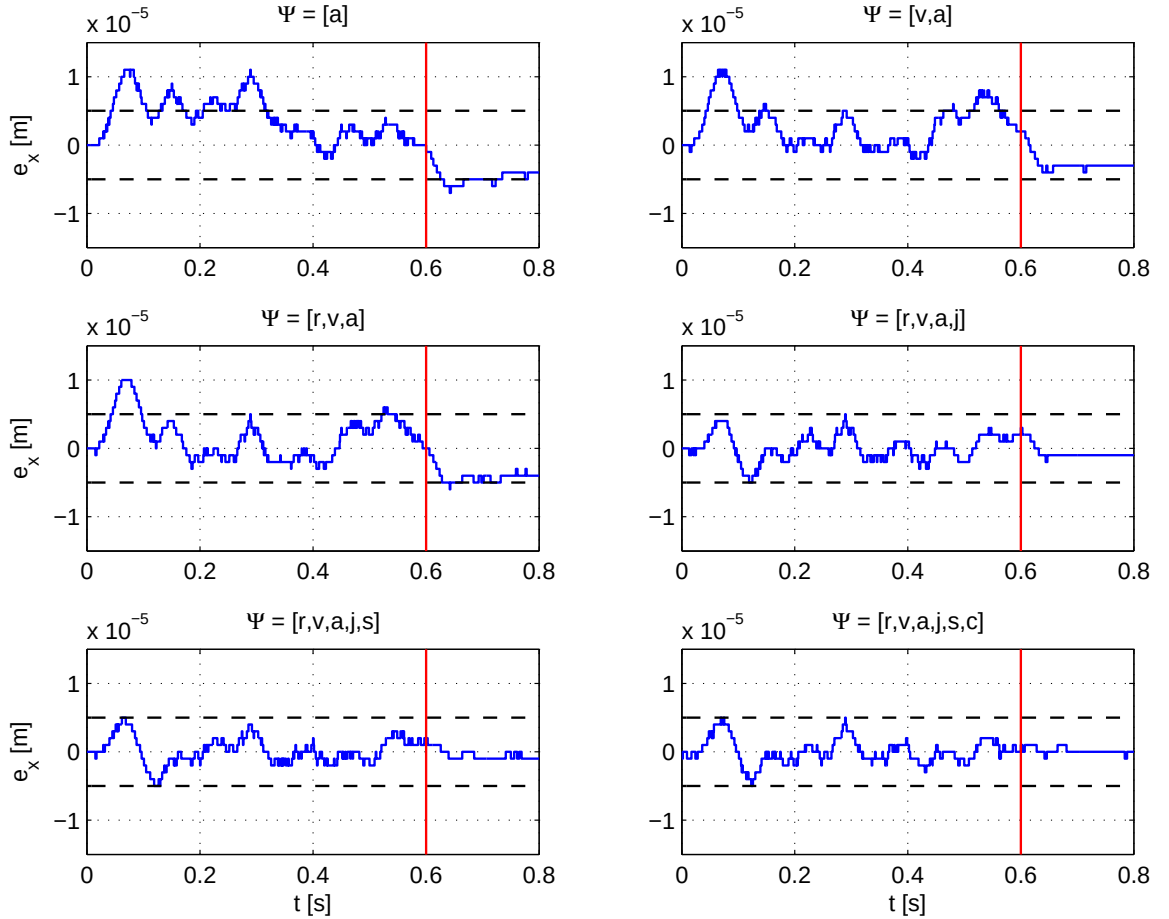


Figure 5.2: The tracking error e_x . The basis functions are introduced step wise: 1. $\Psi = [a]$. 2. $\Psi = [v, a]$. 3. $\Psi = [r, v, a]$. 4. $\Psi = [r, v, a, j]$. 5. $\Psi = [r, v, a, j, s]$. 6. $\Psi = [r, v, a, j, s, c]$.

From Figure 5.2 it is clear that the best tracking performance in x is achieved with basis $\Psi = [r, v, a, j, s, c]$. With this basis the error e_x around the final value of the reference (last 200 ms) is of the same order of magnitude as the sensor accuracy ($1 \mu\text{m}$).

As argued in section 3.2, the accuracy requirements are defined as

$$-5 \mu\text{m} < e_{x_L} < 5 \mu\text{m} \quad (5.3)$$

$$-5 \mu\text{m} < e_{x_R} < 5 \mu\text{m}. \quad (5.4)$$

The errors e_{x_L} and e_{x_R} , the feed forward signal f_x and the input signal u_x for $\Psi = [r, v, a, j, s, c]$ are plotted in Figure 5.3.

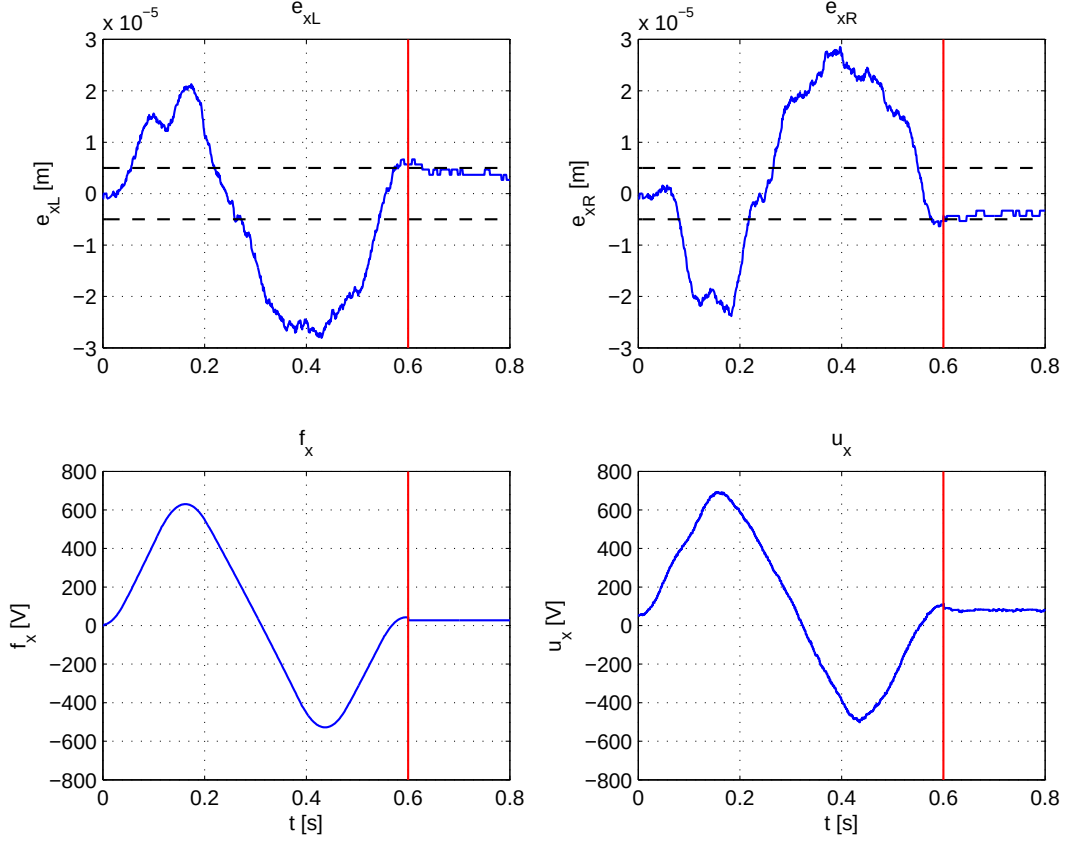


Figure 5.3: The tracking errors e_{x_L} and e_{x_R} , the feed forward signal f_x and the input signal u_x for $\Psi = [r, v, a, j, s, c]$.

The feed forward signal f_x is clearly dominated by the acceleration profile. This is also confirmed by the feed forward parameters θ which correspond to the normalized basis functions of Figure 4.3

$$\theta = \begin{bmatrix} \theta_r \\ \theta_v \\ \theta_a \\ \theta_j \\ \theta_s \\ \theta_c \end{bmatrix} = \begin{bmatrix} 46.7755 \\ 45.3958 \\ 635.6832 \\ 14.0031 \\ 7.3057 \\ -22.3296 \end{bmatrix}. \quad (5.5)$$

Even though the acceleration profile is dominant, the other basis functions have a significant contribution to f_x . This is in accordance with Figure 5.2 where it is clearly shown that the rest of the basis functions significantly contribute to reduce the error in the average position x .

However, in spite of the fact that the accuracy requirements are fully satisfied in the average position x , Figure 5.3 clearly shows that the accuracy requirements are not satisfied at the left and right end of the gantry beam. However, Figure 5.2 shows that the error e_x is in the order of the sensor accuracy. Therefore, it can be concluded that to meet the accuracy requirements in both x_L and x_R , feed forward control in ϕ is necessary.

5.2 MIMO ILC controller

In this section the experimental results related to the MIMO ILC controller are presented. First, in section 5.2.1, the case is considered in which W_e is chosen as in (4.28). Then, in section 5.2.2, W_e is modified in order to improve the tracking performance.

5.2.1 Standard MIMO ILC with basis functions

Section 5.1 showed that the basis $\Psi = [r, v, a, j, s, c]$ results in satisfactory tracking performance in x for the SISO ILC controller. Therefore also for MIMO ILC, Ψ_x is chosen $\Psi_x = [r, v, a, j, s, c]$. The choice of Ψ_ϕ is free and it is not necessary to choose the same basis functions as used in Ψ_x . However, to incorporate extrapolation properties the basis functions should be chosen as a function of the reference. Therefore, also Ψ_ϕ is chosen $\Psi_\phi = [r, v, a, j, s, c]$.

Figure 5.4 shows the errors e_x , e_ϕ , e_{xL} and e_{xR} for $\Psi_x = [r, v, a, j, s, c]$ and $\Psi_\phi = [r, v, a, j, s, c]$.

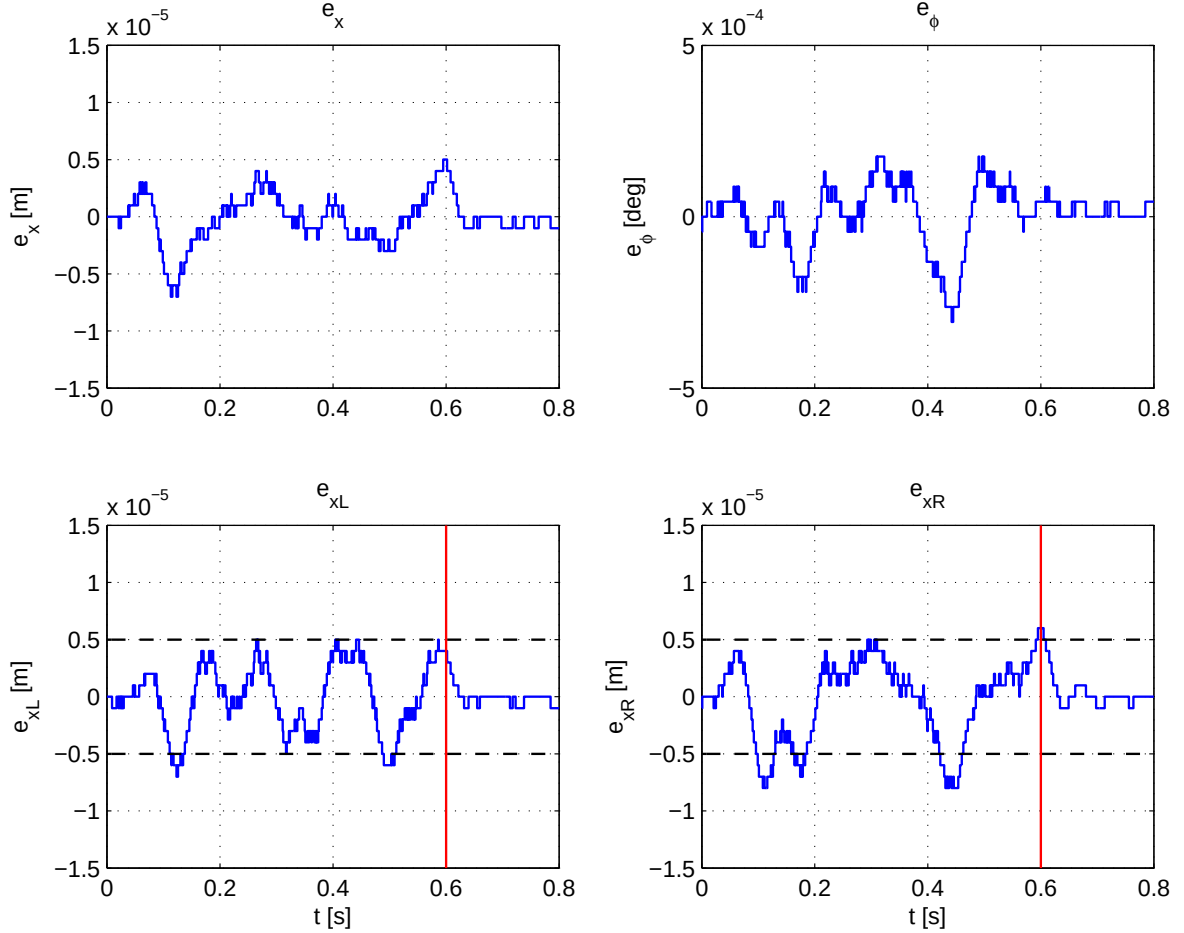


Figure 5.4: e_x , e_ϕ , e_{xL} and e_{xR} for $\Psi_x(r) = [r, v, a, j, s, c]$ and $\Psi_\phi(r) = [r, v, a, j, s, c]$.

It is clear that the accuracy requirements are not satisfied, since e_{xR} at 0.6 s is larger than $5\mu\text{m}$. However, in contrast to SISO ILC, for MIMO ILC the main contribution to the errors e_{xL} and e_{xR} is the error e_x .

5.2.2 MIMO ILC with basis functions with W_e modified

A possible approach to improve the performance is to change the weighting W_e in the cost criterion of (4.9). As concluded from Figure 5.4, e_x has a major contribution to e_{xL} and e_{xR} compared to e_ϕ . Therefore, the weighting on e_x is increased by a factor 25.

As argued in chapter 3, the accuracy requirements should be satisfied when the printing task is started, i.e. in the last 200 ms of the trial. Therefore, W_e can also be modified such that both e_x and e_ϕ have a larger weighting on the final part of the trial compared to the rest of the trial. To this end, the last 500 diagonal elements of W_e have a value 2 times larger than the rest of the diagonal elements. The values presented here are a result of an iterative-tuning procedure. The results of both modifications on W_e are shown in Figure 5.5 and the feed forward signals f_x and f_ϕ and the input signals u_x and u_ϕ are plotted in Figure 5.6.

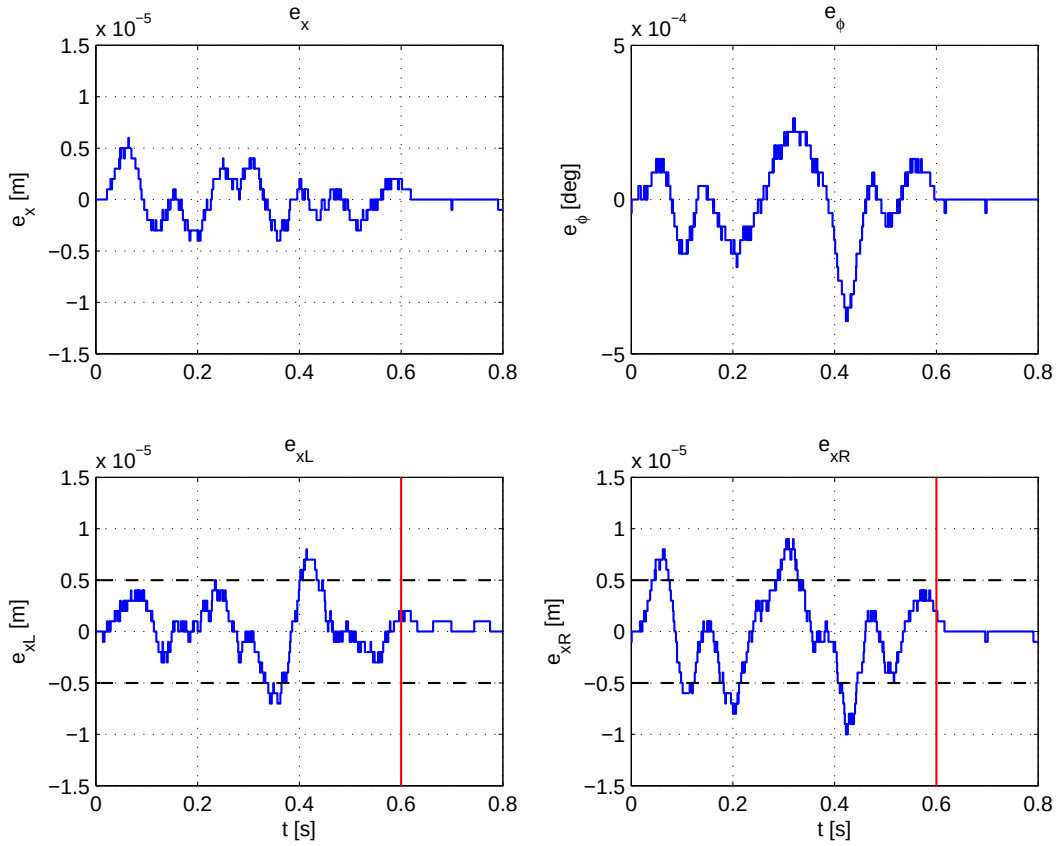


Figure 5.5: e_x , e_ϕ , e_{xL} and e_{xR} for $\Psi_x(r) = [r, v, a, j, s, c]$ and $\Psi_\phi(r) = [r, v, a, j, s, c]$ with W_e modified.

Figure 5.5 shows that during the last 200 ms of the trial, both e_{xL} and e_{xR} are of the same order of magnitude as the sensor accuracy ($1 \mu\text{m}$). Therefore, it can be concluded that the designed MIMO ILC controller with modified W_e satisfies the accuracy requirements.

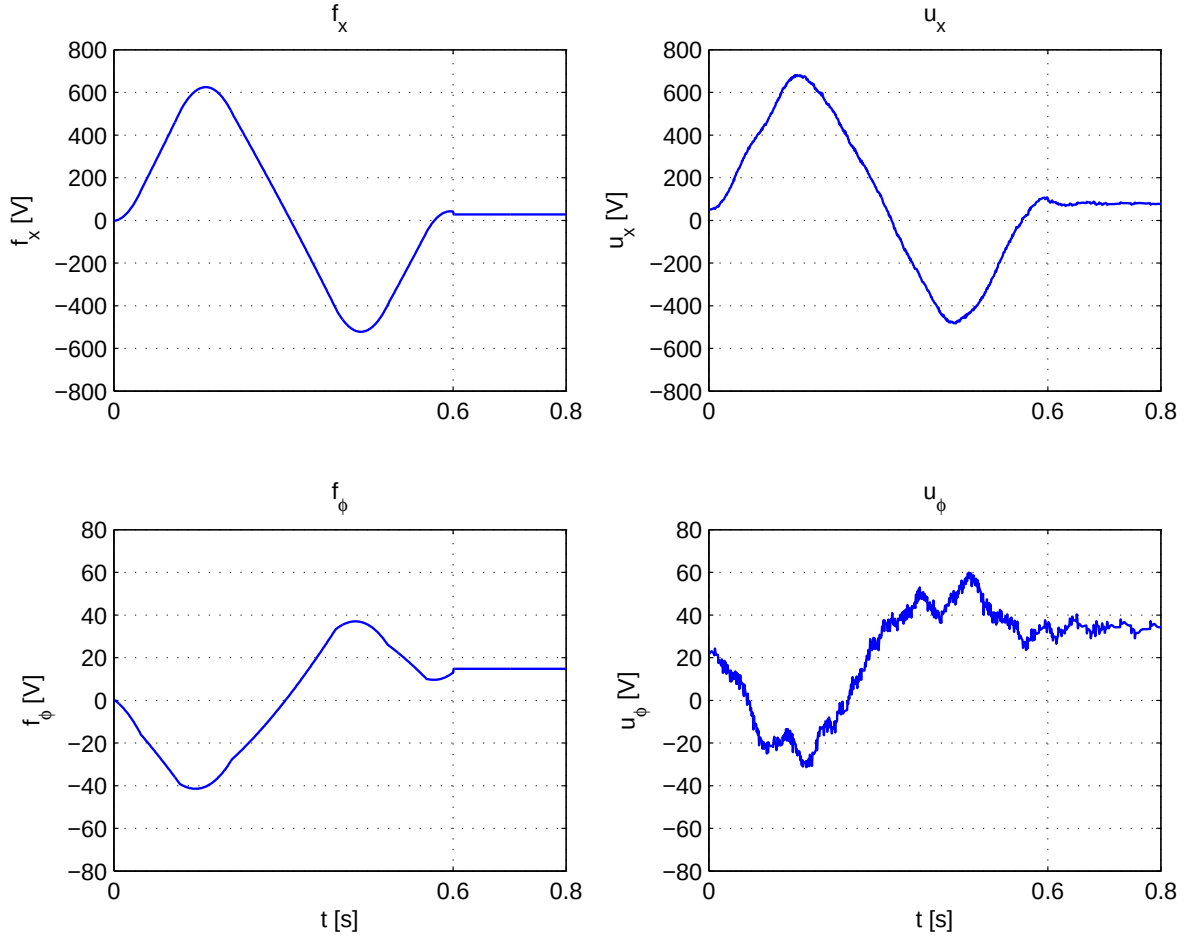


Figure 5.6: f_x , f_ϕ , u_x and u_ϕ for $\Psi_x(r) = [r, v, a, j, s, c]$, $\Psi_\phi(r) = [r, v, a, j, s, c]$ with W_e modified.

Figure 5.6 shows that also for the MIMO ILC controller the signals f_x and f_ϕ are dominated by the acceleration profile as confirmed by (5.6). It should be noted that the rest of the feed forward parameters are of significant order. Therefore, to achieve the required accuracy all the considered basis functions are of crucial importance.

$$\theta_x = \begin{bmatrix} \theta_{x,r} \\ \theta_{x,v} \\ \theta_{x,a} \\ \theta_{x,j} \\ \theta_{x,s} \\ \theta_{x,c} \end{bmatrix} = \begin{bmatrix} 37.6715 \\ 57.4345 \\ 622.2424 \\ 19.9577 \\ -4.5971 \\ -9.7235 \end{bmatrix}, \theta_\phi = \begin{bmatrix} \theta_{\phi,r} \\ \theta_{\phi,v} \\ \theta_{\phi,a} \\ \theta_{\phi,j} \\ \theta_{\phi,s} \\ \theta_{\phi,c} \end{bmatrix} = \begin{bmatrix} 16.7596 \\ -7.8840 \\ -36.0215 \\ -6.6616 \\ -1.9143 \\ 3.3164 \end{bmatrix}. \quad (5.6)$$

5.3 Extrapolation properties

In this section the extrapolation properties of both ILC controllers are investigated. To this end, the set of references introduced in section 3.1 is used. These references are used in an experiment consisting of 30 trials. Trial 1 to 10 are performed on r_{20} , trial 11 to 20 are performed on r_{41} and trial 21 to 30 are performed on r_{61} . Figure 5.7 shows the tracking performance $J = e^T W_e e$, with e the MIMO error consisting of both e_x and e_ϕ for both the SISO ILC controller and the MIMO ILC controller.

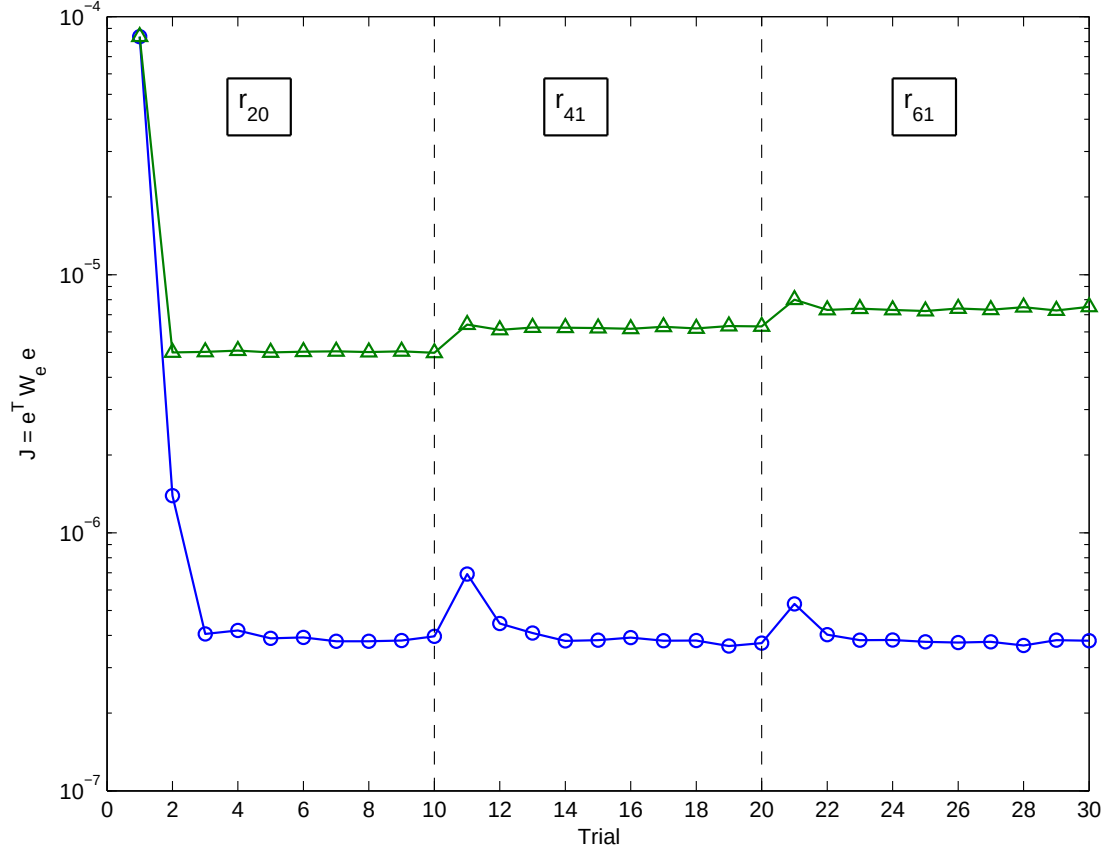


Figure 5.7: Tracking performance $J = e^T W_e e$ for the MIMO ILC controller ($\circ$) and the SISO ILC controller (Δ)

Figure 5.7 shows that both ILC controllers are almost insensitive to a change in reference since the tracking performance J remains of the same order of magnitude as before the change of reference. Figure 5.7 also confirms the difference in performance between the SISO ILC controller and the MIMO ILC controller due to the fact that e_ϕ is not reduced with the SISO ILC controller.

Figure 5.8 shows the tracking errors e_{xL} and e_{xR} for trial 10 to 12 and for trial 20 to 22 of the MIMO ILC controller.

Comparing trial 10 with trial 11 and trial 20 with trial 21 shows that changing the reference does change the tracking performance, especially when the reference is changed from r_{41} to r_{61} . Despite the change of tracking performance, the accuracy requirements are still satisfied.

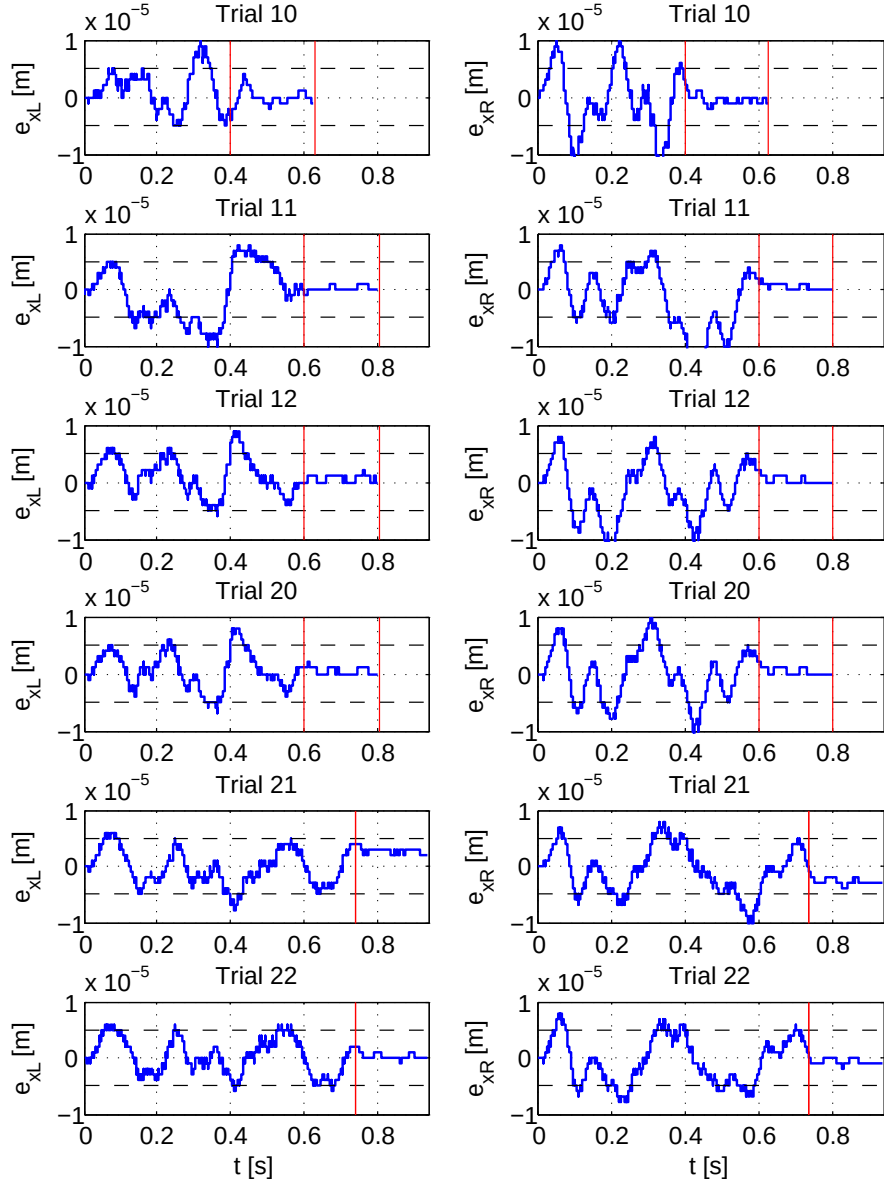


Figure 5.8: The tracking errors e_{xL} and e_{xR} for the MIMO ILC controller for trial 10 to 12 and for trial 20 to 22. Left column: e_{xL} , right column e_{xR}

Comparing trial 11 with trial 12 and trial 21 with trial 22 shows that when one trial is used to learn the new reference a tracking error of the same order of magnitude as the sensor accuracy ($1 \mu\text{m}$) is reached after this 'extra' trial.

It can therefore be concluded that the designed MIMO ILC controller achieves satisfactory tracking performance for the entire set of references confirming the extrapolation properties of ILC with basis functions.

Chapter 6

Conclusions and recommendations

In this work, a feed forward control strategy is presented that aims to track aggressive motion profiles for the gantry beam of the Arizona flatbed printer in order to increase the productivity. Since manual tuning of feed forward controllers is a time consuming process, this work focuses on data-driven auto-tuning feed forward control. To this end, iterative learning control (ILC) with basis functions is considered.

Both a SISO ILC controller as well as a MIMO ILC controller are designed. The accuracy requirements consist of $\pm 5 \mu\text{m}$ error bounds around the end position of the left and right end of the gantry beam. The experimental results show that the designed MIMO ILC controller satisfies the accuracy requirements, in contrast to the SISO ILC controller which only achieves satisfactory performance in the average position x in the center of the gantry beam.

The Arizona flatbed printer executes different printing modes resulting in different reference trajectories for the gantry beam. Therefore, extrapolation properties are essential to guarantee satisfactory performance. To investigate the extrapolation properties of the MIMO ILC controller a set of references is considered consisting of three different motion profiles. Experimental results show that the designed MIMO ILC controller with basis functions achieves good tracking performance for the entire set of motion profiles.

In this work the situation is considered with the carriage at a fixed position. However, to further improve the productivity of the printer current research is towards the situation in which the gantry beam performs its motion task while the carriage is moving. To this end, it is recommended to extend the MIMO structure presented in this work with the interaction between carriage and gantry beam.

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Bibliography

- [1] Van der Meulen, S., Tousain, R., Bosgra, O., 2007. Fixed structure feedforward controller tuning exploiting iterative trials, applied to a high-precision electromechanical servo system. Proceedings of the 2007 American Control Conference, 4033-4039, New York City, New York, USA.
- [2] Bristow, D., Tharayil, M., Alleyne, A., 2006. A survey of iterative learning control: a learning-based method for high performance tracking control. IEEE Control Systems Magazine 26, 96-114.
- [3] Bolder, J., Oomen, T., Koekebakker, S., Steinbuch, M. 2014. Using iterative learning control with basis functions to compensate medium deformation in a wide-format inkjet printer. Mechatronics 3, 2014.
- [4] Bolder, J., Lemmen, B., Koekebakker, S., Oomen, T., Bosgra, O., Steinbuch, M. 2012. Iterative learning control with basis functions for media positioning in scanner inkjet printers. IEEE Multi-Conference on Systems and Control, Dubrovnik, Croatia.
- [5] Steinbuch, M., Van de Molengraft, R. 2000. Iterative learning control of industrial motion systems. Proceedings of the 2000 IFAC conference, 967-972, Darmstadt, Germany.
- [6] Gunnarsson, S., Norröf, M. 2001. On the design of ILC algorithms using optimization. Automatica 37, 2011-2016.
- [7] Janssens, P., Pipeleers, G., Swevers, J. 2013. A Data-Driven Constrained Norm-Optimal Iterative Learning Control Framework for LTI Systems. IEEE Transactions on Control Systems Technology 21, no 2, 546-551.
- [8] Van de Wijdeven, J., Bosgra, O. 2010. Using basis functions in iterative learning control: analysis and design theory. International Journal of Control 83, no. 4, 661-675.
- [9] Lepelaars, D., Koekebakker, S., Severt, R., Steinbuch, M. 2014. Low cost linear stepper design through smart encoder system. Master's thesis. Eindhoven University of Technology, The Netherlands.
- [10] Skogestad, S., Postlethwaite, I. 2005. Multivariable feedback control: Analysis and design. New York: John Wiley & Sons, Inc.